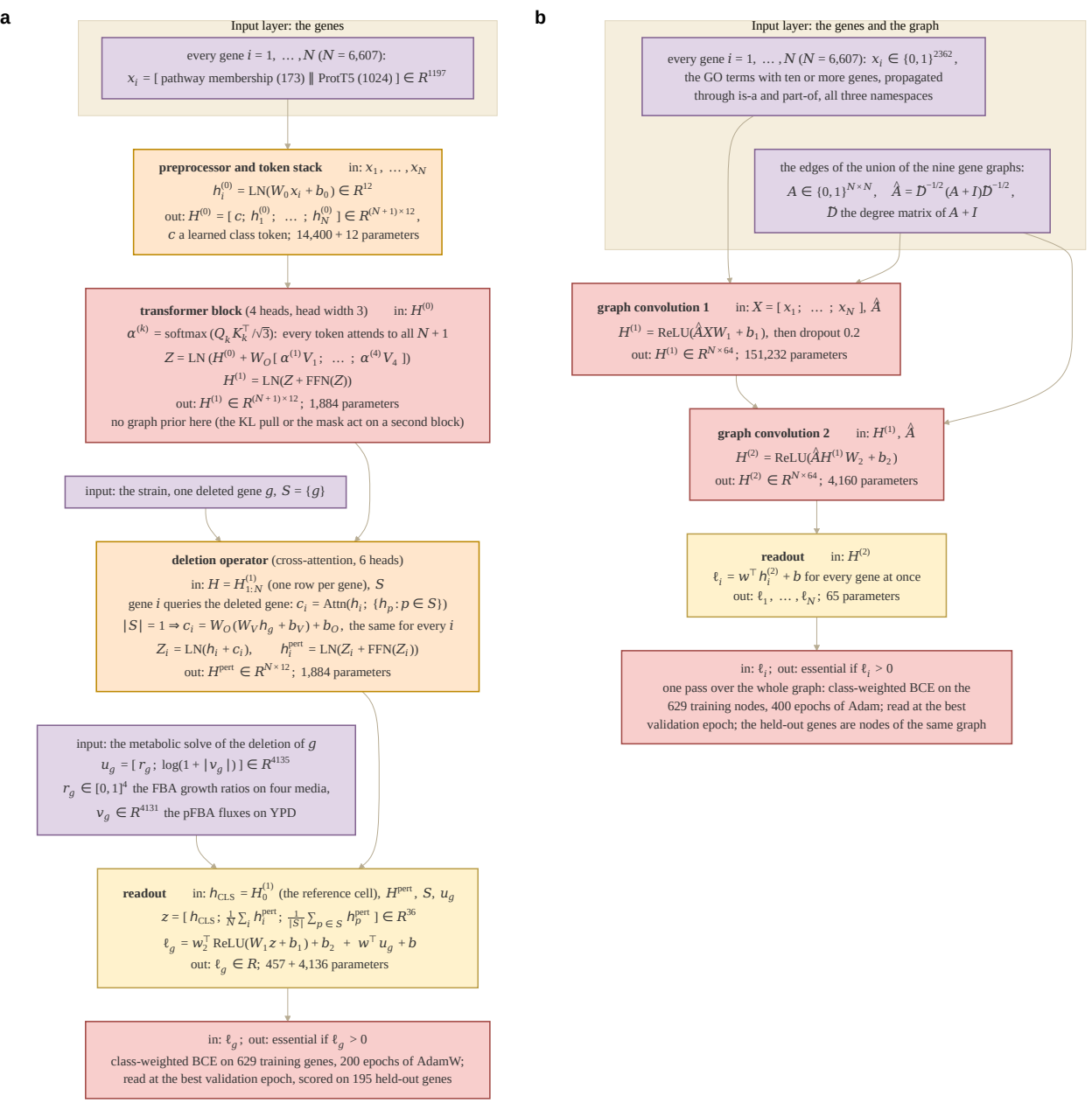

a
Input layer: the genes
every gene $i = 1, \dots, N$ (N = 6,607):
x_i = [pathway membership (173) ‖ ProtT5 (1024)] $\in R^{1197}$
preprocessor and token stack in: $x_1, \dots, x_N$
$h_i^{(0)} = \mathrm{LN}(W_0 x_i + b_0) \in R^{12}$
out: $H^{(0)} = [\, c;\ h_1^{(0)};\ \dots;\ h_N^{(0)} \,] \in R^{(N+1)\times 12}$,
c a learned class token; 14,400 + 12 parameters
transformer block (4 heads, head width 3) in: $H^{(0)}$
$\alpha^{(k)} = \mathrm{softmax}\,(Q_k K_k^\top/\sqrt{3})$: every token attends to all $N+1$
$Z = \mathrm{LN}\,(H^{(0)} + W_O\,[\, \alpha^{(1)} V_1;\ \dots;\ \alpha^{(4)} V_4 \,])$
$H^{(1)} = \mathrm{LN}(Z + \mathrm{FFN}(Z))$
out: $H^{(1)} \in R^{(N+1)\times 12}$; 1,884 parameters
no graph prior here (the KL pull or the mask act on a second block)
input: the strain, one deleted gene g, $S = \{g\}$
deletion operator (cross-attention, 6 heads)
in: $H = H^{(1)}_{1:N}$ (one row per gene), S
gene i queries the deleted gene: $c_i = \mathrm{Attn}(h_i;\ \{h_p : p \in S\})$
$|S| = 1 \Rightarrow c_i = W_O(W_V h_g + b_V) + b_O$, the same for every i
$Z_i = \mathrm{LN}(h_i + c_i)$, $h_i^{\mathrm{pert}} = \mathrm{LN}(Z_i + \mathrm{FFN}(Z_i))$
out: $H^{\mathrm{pert}} \in R^{N\times 12}$; 1,884 parameters
input: the metabolic solve of the deletion of g
$u_g = [\, r_g;\ \log(1 + |v_g|) \,] \in R^{4135}$
$r_g \in [0,1]^4$ the FBA growth ratios on four media,
$v_g \in R^{4131}$ the pFBA fluxes on YPD
readout in: $h_{\mathrm{CLS}} = H_0^{(1)}$ (the reference cell), H^{pert}, S, u_g
$z = [\, h_{\mathrm{CLS}};\ \frac{1}{N}\sum_i h_i^{\mathrm{pert}};\ \frac{1}{|S|}\sum_{p\in S} h_p^{\mathrm{pert}} \,] \in R^{36}$
$\ell_g = w_2^\top \mathrm{ReLU}(W_1 z + b_1) + b_2 \ +\ w^\top u_g + b$
out: $\ell_g \in R$; 457 + 4,136 parameters
in: ℓ_g; out: essential if $\ell_g > 0$
class-weighted BCE on 629 training genes, 200 epochs of AdamW;
read at the best validation epoch, scored on 195 held-out genes
b
Input layer: the genes and the graph
every gene $i = 1, \dots, N$ (N = 6,607): $x_i \in \{0,1\}^{2362}$,
the GO terms with ten or more genes, propagated
through is-a and part-of, all three namespaces
the edges of the union of the nine gene graphs:
$A \in \{0,1\}^{N\times N}$, $\hat{A} = \hat{D}^{-1/2}(A + I)\hat{D}^{-1/2}$,
$\hat{D}$ the degree matrix of $A + I$
graph convolution 1 in: $X = [\, x_1;\ \dots;\ x_N \,]$, $\hat{A}$
$H^{(1)} = \mathrm{ReLU}(\hat{A}XW_1 + b_1)$, then dropout 0.2
out: $H^{(1)} \in R^{N\times 64}$; 151,232 parameters
graph convolution 2 in: $H^{(1)}$, $\hat{A}$
$H^{(2)} = \mathrm{ReLU}(\hat{A}H^{(1)}W_2 + b_2)$
out: $H^{(2)} \in R^{N\times 64}$; 4,160 parameters
readout in: $H^{(2)}$
$\ell_i = w^\top h_i^{(2)} + b$ for every gene at once
out: $\ell_1, \dots, \ell_N$; 65 parameters
in: ℓ_i; out: essential if $\ell_i > 0$
one pass over the whole graph: class-weighted BCE on the
629 training nodes, 400 epochs of Adam; read at the best
validation epoch; the held-out genes are nodes of the same graph